

1 introduction

The baseline model has 34.826 parameters. Is it possible to obtain the same or better performance on the MNIST digit dataset by reducing the parameters to under 7.000? This overview is a summary of the process used to reduce the parameters and assess the results by observing the model's accuracy. The architecture of the baseline model is shown in Table 1. The baseline model accuracy is $\sim 98.75\%$.

Layer Type	Configuration	Kernel Size
Conv2D	32 filters	3×3
MaxPool2D	-	2×2
Conv2D	64 filters	3×3
MaxPool2D	-	2×2
Flatten	-	-
Dropout	rate = 0.5	-
Dense (Output)	10 units (Softmax)	-

Table 1: Baseline model architecture overview

2 introducing the first model

Given the low complexity of the dataset, which consists of images containing single digits, the parameter reduction strategy initially focused on decreasing the number of filters in the convolutional layers and adding a dense layer before the output layer to compress information. Furthermore, Max Pooling was replaced by Average Pooling to preserve more contextual information because the images in the dataset do not have sharp edges. Table 2 shows the architecture of the new model. The first proposed model

Layer Type	Configuration	Kernel Size
Conv2D	8 filters	3×3
AveragePool2D	-	2×2
Conv2D	16 filters	3×3
AveragePool2D	-	2×2
Flatten	-	-
Dropout	rate = 0.5	-
Dense	8 units (ReLU)	-
Dense (Output)	10 units (Softmax)	-

Table 2: First model architecture overview.

have 4.546 parameters and accuracy of $\sim 97.3\%$.

3 Introducing the second model

The second model has an architecture similar to the model proposed in advance, but with two differences: the first is the kernel size in the first convolutional layer, the second is the removal of the first dense layer. Table 3 shows this new architecture. The fil-

Layer Type	Configuration	Kernel Size
Conv2D	8 filters	7×7
AveragePool2D	-	2×2
Conv2D	16 filters	3×3
AveragePool2D	-	2×2
Flatten	-	-
Dropout	rate = 0.4	-
Dense (Output)	10 units (Softmax)	-

Table 3: Second model architecture overview.

ter's number in the convolutional layer is the same as the model in Table 2 but the first convolutional layer in the second model has a kernel size of 7×7 . The dimension of the filter was increased trying to capture low level feature by increasing the perceptive field of the filter and reduce the parameters of the second convolutional layer. Now the model can process more structured data in the first layer and try to compensate the parameters reduction. After this the first dense layer was removed for two reasons: first, it introduce a lot of parameters; second, the task is very simple and a second layer may introduce complexity. At the end the dropout was decreased to 0.4. This strategy worked well, model parameters are reduced to 4.138 and the accuracy of the second model is $\sim 98.92\%$.

4 Introducing the third model

The third model represents a shift in approach compared to the previous. Whereas the second model utilized a larger kernel size in the first layer to reduce parameters, this model employs a consistent 3×3 kernel size across all convolutional layers, introduces an additional layer with 8 filters and removes the final pooling layer after the last convolution. This was necessary to preserve the spatial structure of the information, avoiding its premature reduction to a vector. Table 4 shows the new architecture that achieves an

Layer Type	Configuration	Kernel Size
Conv2D	8 filters	3×3
AveragePool2D	-	2×2
Conv2D	8 filters	3×3
AveragePool2D	-	2×2
Conv2D	16 filters	3×3
Flatten	-	-
Dropout	rate = 0.3	-
Dense (Output)	10 units (Softmax)	-

Table 4: Fourth model architecture overview.

accuracy of $\sim 99.02\%$. This sets the third model, with 3.282 parameters, as the best model. The reduction strategy worked well, the accuracy is the same as the baseline but the complexity is reduced to one-tenth.